

SCIENCE FAIR REPORT

Project Title: Renewable smart home model

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Problem statement:

Today most homes still depend on electricity from fossil fuels and water supplied through large systems, which are expensive, unreliable, and harmful to the environment. Power shortages, rising energy costs, and water wastage are common problems in developing countries. As technology continues to advance, the homes in the future will no longer be simple buildings. They will have smart systems such as voice-controlled lighting, automatic energy management, and intelligent water usage.

However, many smart homes still rely on non-renewable energy, which limits their sustainability. There is a need to show how renewable energy and smart technology can work together. This project addresses that need by creating a model of a smart home that uses clean energy, collects rainwater, and controls lighting automatically to demonstrate a more sustainable and efficient future.

Hypothesis:

If a home is powered by renewable energy and uses smart systems such as automatic and voice-controlled lighting, energy management, and rainwater harvesting, then it will use less electricity and water while reducing environmental impact. This approach will also help foster technological advancement and positive change by encouraging the development and adoption of smarter, more sustainable homes for the future.

Scope:

- A miniature house model
- renewable energy system
- A rechargeable battery to store energy
- Automatic lighting/ smart lighting
- Rainwater harvesting
- led lights to demonstrate the operation of the house

Objectives:

- To design and build a working model of a smart home powered by at least 1 renewable energy source such as solar power.
- To demonstrate how renewable energy can be generated, stored, and used efficiently in a home.
- To develop an automatic lighting system using sound and light sensors to reduce unnecessary electricity use and make the home smart.
- To design a rainwater harvesting and storage system that shows how water can be collected, filtered and reused instead of wasted.
- To show how smart technology can be implemented in modern homes.
- To demonstrate how future homes can combine technology and environmental responsibility.

Materials:

Structure

- Large wooden board
- Cardboard
- Glue
- Construction set
- Stationary
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Energy system

- Small 5v solar panels
- rechargeable batteries+holder
- Connecting wires

Smart system

- Arduino microprocessor + cable
- Jumper wires
- Sound sensor
- Leds
- Switches
- LDR

Rainwater harvesting system

- Plastic tubes
- Mesh filter
- Plastic container

Safety precautions:

- Only low voltage components will be used
- No connection to mains power supply
- Correct use of well insulated wires and connections

Conclusion: This project will demonstrate how renewable energy and smart automation can be used together to create a sustainable home of the future. It will help people understand how technology can reduce environmental impact and improve daily life.